

Adaptive Cyber Threat Intelligence under Concept Drift

Mini Project Concept — Advanced Machine Learning

1. Core Idea

Instead of building a cyber-threat model once and keeping it fixed, the proposed system continuously monitors changing network behaviour. When the relationship between network features and threat behaviour changes (concept drift), the system identifies the change and adapts or retrains the ML model using newer data.

2. Input and Output

- Input: Network-flow records containing features such as protocol, destination port, flow duration, packet/byte counts, packet length, flow rate and TCP flags.
- Threat output: Classification such as Normal, DoS, Brute Force, Botnet or Port Scan, depending on the selected dataset.
- Confidence output: Probability/confidence associated with the predicted class.
- Adaptation output: Model status such as Stable or Concept Drift Detected → adaptation/retraining required.

3. How the System Works

1. **Network Traffic:** Collect historical and incoming network-flow records.
2. **Preprocessing:** Clean data, encode categorical values and scale features where required.
3. **Feature Selection:** Use methods such as RFE or Mutual Information to retain useful features.
4. **ML Modelling:** Train and compare Logistic Regression (baseline), SVM and XGBoost/LightGBM.
5. **Monitoring:** Evaluate incoming time windows and monitor changes in data/model behaviour.
6. **Drift Detection:** Identify whether the underlying traffic pattern has changed significantly.
7. **Adaptation:** If drift is detected, update/retrain the model using recent data and re-evaluate it.
8. **Explainability:** Use SHAP to show which network features influenced important predictions.

4. Advanced ML Concepts We Can Use

- Advanced supervised learning: SVM with kernels and gradient-boosting models such as XGBoost/LightGBM.
- Ensemble learning: compare boosting and, if useful, stacking/blending.
- Hyperparameter tuning: Random Search or Bayesian Optimization.
- Feature selection: RFE / Mutual Information.
- Model interpretability: SHAP (and optionally LIME).
- Advanced evaluation: Precision, Recall, F₁-score, Precision–Recall Curve and ROC-AUC.
- ML systems: Concept Drift, Model Monitoring and adaptive/periodic model updating.
- Optimization: SGD-based updating can be explored if an incremental-learning approach is selected.

5. Simplified Architecture

Network Traffic → Preprocessing → Feature Selection → SVM / XGBoost / LightGBM
→ Threat Classification + SHAP Explanation → Model Monitoring → Concept Drift?
→ No: Continue Monitoring | Yes: Adapt/Retrain → Updated Model → Continue Monitoring

6. Why This Is an Advanced ML Problem

The project is not simply about detecting attacks. Its main research problem is maintaining reliable ML performance when network behaviour changes over time. This gives a natural reason to use concept drift, model monitoring, adaptive updating, ensemble learning, hyperparameter optimization, feature selection, explainability and advanced evaluation rather than adding these techniques artificially.

7. Possible Mini-Project Problem Statement

Traditional cyber-threat ML models are often trained on historical network behaviour and may lose effectiveness when traffic patterns and attack behaviours evolve. The proposed project aims to develop an adaptive machine-learning framework that monitors network traffic for concept drift, evaluates model degradation, and updates the model using recent data while providing interpretable threat predictions.

8. Suggested Model Set

Baseline: Logistic Regression **Advanced models:** SVM + XGBoost/LightGBM **Explainability:** SHAP

Adaptation: Concept Drift + Model Monitoring + Retraining/Updating